R13

Code No:111AD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD
B.Tech I Year Examinations, December - 2018
ENGINEERING PHYSICS
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, ETM, MMT, AE, AME, MIE, PTM, MSNT, AGE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is have a, b, c as sub questions. compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may

PART - A

(25 Marks)

1.a)	Define atomic radius, co-ordination number?	[2]
b)	Draw the following planes of cubic structure. (121), (010), (202)	[3]
c)	State and explain Heisenberg uncertainty principle.	[2]
d)	What do you understand from Davisson and Germer experiment?	[3]
e)	Define Electric dipole, polarizability.	[2]
f)	Define piezoelectricity and pyroelectricity.	[3]
g)	What is meant by optical resonator?	[2]
h)	How will you classify the optical fibers?	[3]
i)	Origin of nanoscience - explain.	[2]
j)	Explain the working principle of a Solar cell?	[3]

PART-B

(50 Marks)

2.a) Explain the terms (i) basis (ii) space lattice (iii) unit cell.
b) Show that FCC crystals are closely packed than BCC crystals. [5+5]

OR

3.a) Describe the Laue method of determination of crystal structure.
b) Calculate the Bragg angle at which electrons accelerated from rest through a potential difference of 80 volts will be diffracted from the (111) planes of a FCC crystal of lattice parameter 0.35 nm. [5+5]

4.a) Derive Schrodinger time independent wave equation.
b) Explain how Davission-Germer experiment is used to explain the existence of matter waves. [5+5]

OR

5.a) Derive an expression for energy levels of particle enclosed in one dimensional Potential box.
b) How band theory of solids leads to classification of solids in to conductors, semi conductors and insulators. [5+5]

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6.a) Derive an expression for ionic polarizability.
b) Derive Clausius - Mossotti relation. [5+5]

OR

7.a) Discuss the properties and effect of external field on the para and ferro magnetic materials.
b) What is meant by Hysteresis loss? Describe the formation of hysteresis loop using domain wall movement. [5+5]

8.a) What is the difference between interference and diffraction?
b) Briefly explain the N-slits of diffraction pattern. [5+5]

OR

9.a) Why the population inversion is necessary to achieve lasing action?
b) Describe the construction and working of Helium-Neon laser. [5+5]

10.a) Explain the formation of PN junction diode and V-I characteristics of the same.
b) Derive an expression for density of electrons in intrinsic semiconductors. [5+5]

OR

11.a) How do you synthesize the nonmaterial using Chemical Vapor Deposition (CVD) method?
b) Write short note on Ball milling method. [5+5]

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